

Model Performance Overview

INTRODUCTION

In this study, we evaluated six state-of-the-art synthetic tabular data generators, MedGAN, CTGAN, TabDDPM, PATEGAN, CODI, and GANBLR, to understand their ability to reproduce real dataset structure and support downstream machine learning tasks. Each model represents a different generative paradigm, including GAN-based architectures, diffusion models, privacy-preserving frameworks, Bayesian-inspired methods, and hybrid learning systems.

To assess their effectiveness, we used the Train on Synthetic, Test on Real (TSTR) evaluation strategy across five benchmark datasets: CAR, MAGIC, Nursery, Shuttle, and Adult. For every synthetic dataset generated, we trained four classifiers, Logistic Regression, MLP, Random Forest, and XGBoost, and compared their accuracy, F1, and AUC scores against real-data performance.

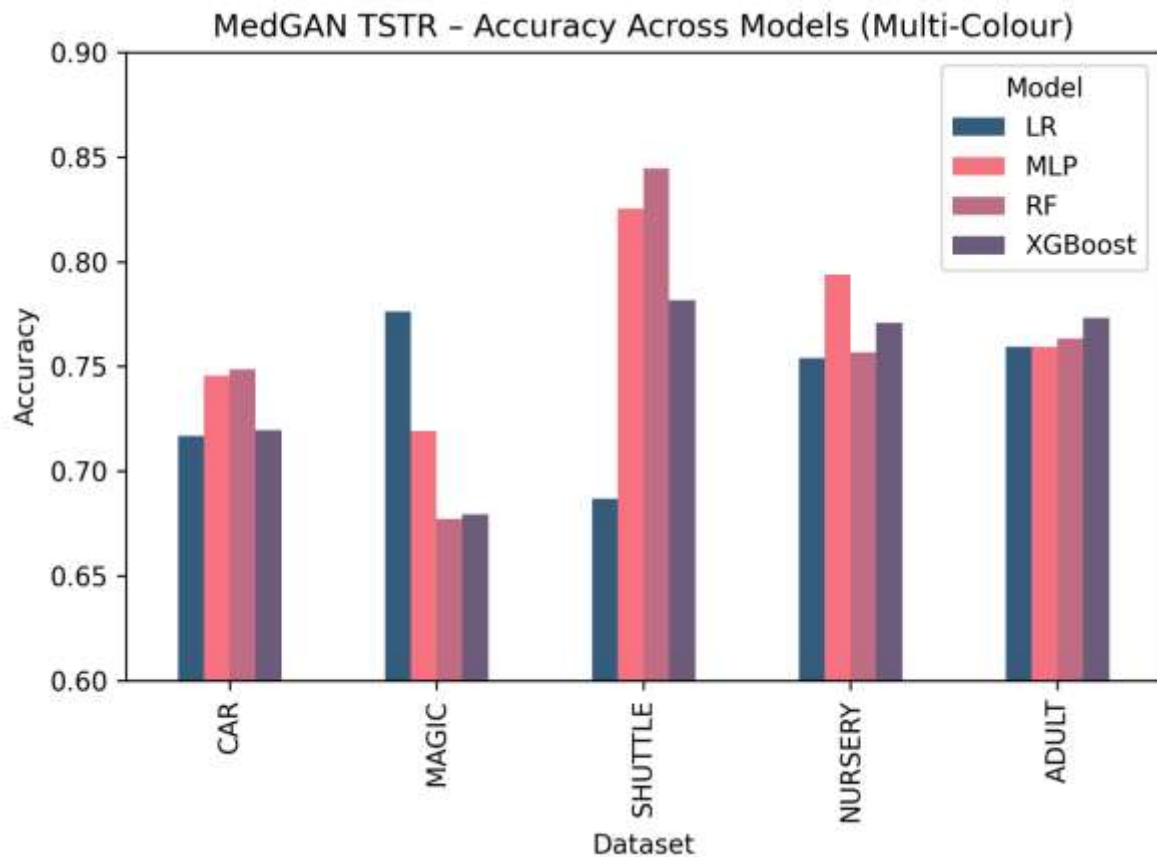
This systematic evaluation allows us to clearly understand which generative model works best for which type of dataset, how model behaviour changes across categorical vs numerical features, and which model provides the most reliable synthetic data quality for downstream ML tasks.

1. MedGAN- Model Details & Performance Summary

MedGAN showed **strong and stable performance**, especially on the **Shuttle** dataset where it achieved very high accuracy and F1 across multiple classifiers. This indicates that MedGAN learns structured numerical patterns very effectively. It also performed well on **Nursery** and **Adult**, showing consistent accuracy and good generalisation.

However, its performance on **Magic** was slightly lower compared to other models, which suggests MedGAN may struggle when feature interactions are more complex or noisy.

Overall, MedGAN is **reliable on well-structured datasets** and provides **balanced results across models**, making it a strong baseline generator.

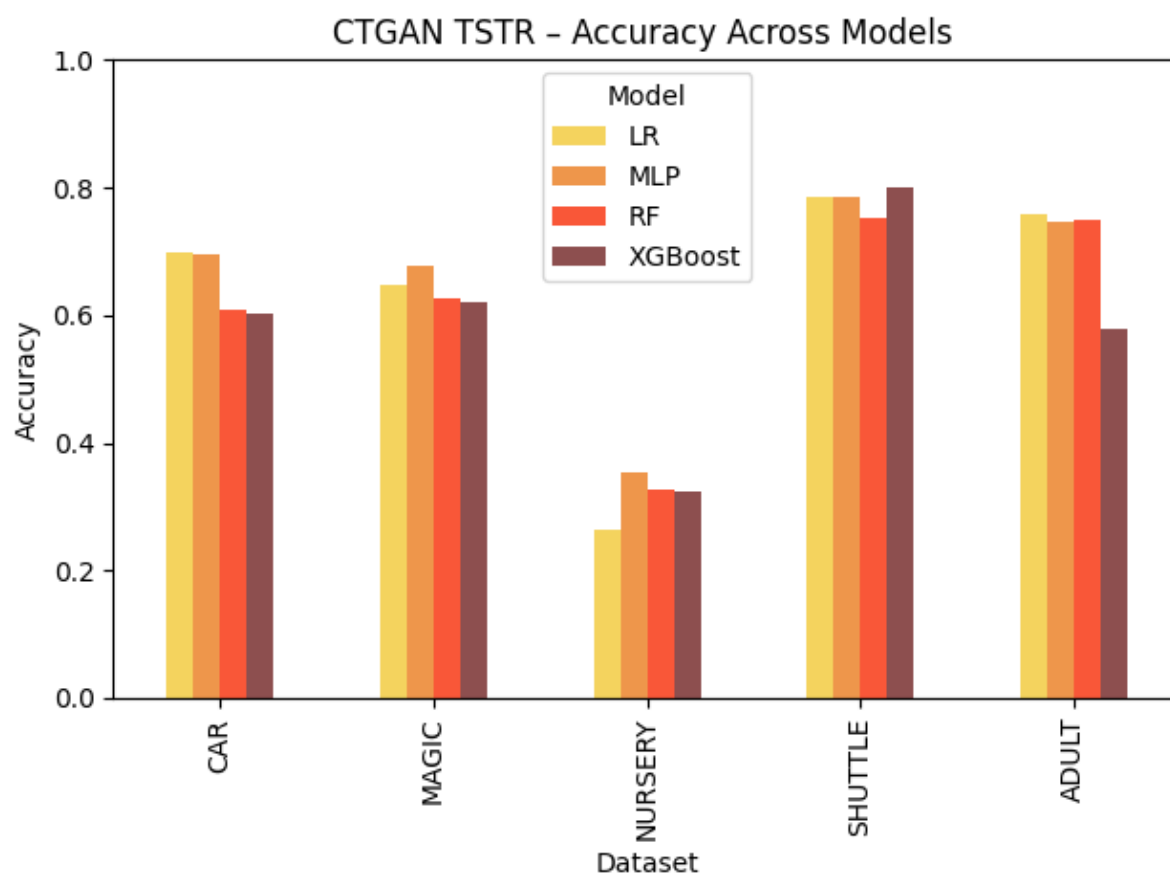


2. CTGAN- Model Details & Performance Summary

CTGAN showed **mixed performance** depending on the dataset. It performed best on **Shuttle** and **Adult**, where Logistic Regression and XGBoost reached competitive accuracy. CTGAN handles continuous and mixed-type features quite well, especially when the dataset has meaningful separation between classes.

However, CTGAN struggled significantly on **highly categorical datasets** like **Nursery** and **CAR**, where the accuracy values dropped. This is expected because CTGAN is more suited for continuous distributions and may fail to capture complex categorical dependencies.

Overall, CTGAN works **best on numeric or mixed datasets**, but is **less effective** for datasets with many categorical levels.

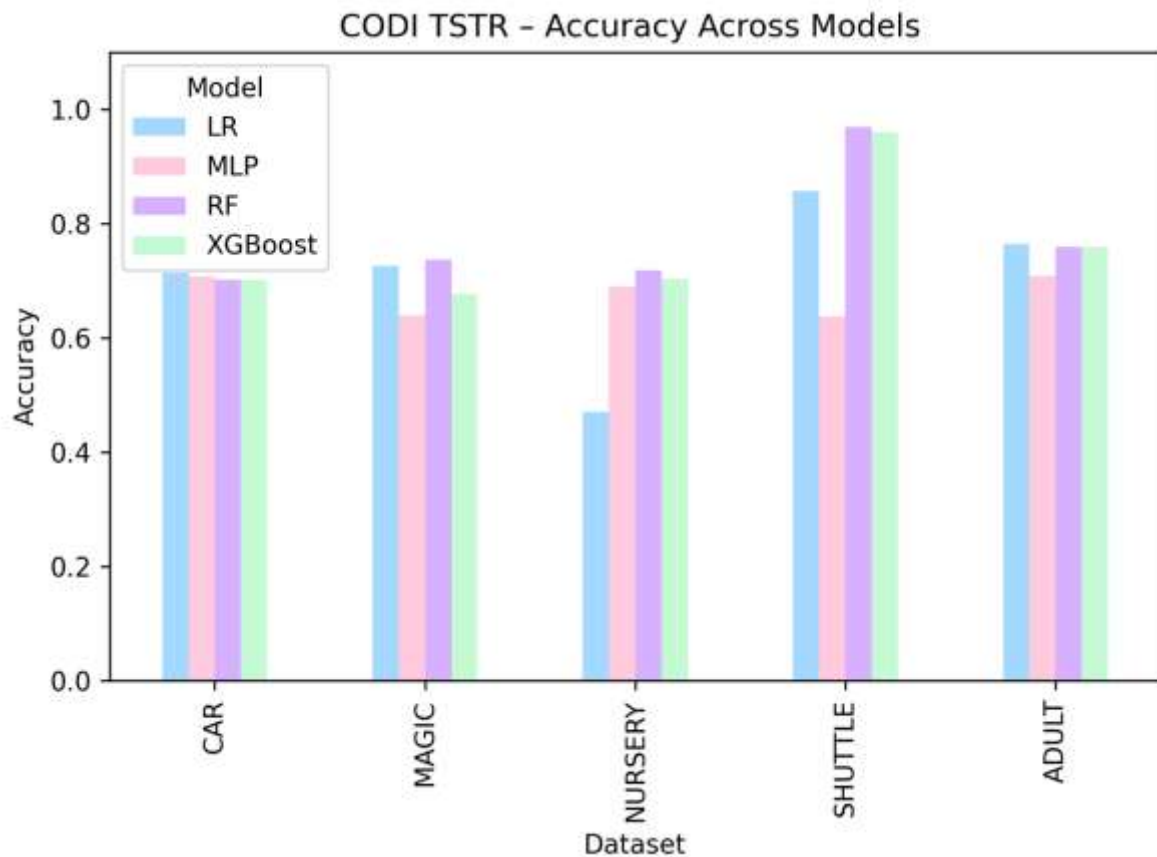


3. CODI- Model Details & Performance Summary

CODI showed **exceptional performance**, especially on the **Shuttle** dataset where Random Forest achieved extremely high accuracy (0.96+) and F1 (>0.94). This indicates that CODI is highly effective at capturing complex numerical relationships.

It also performed strongly on **Magic** and **Nursery**, demonstrating solid handling of both numerical and categorical features. CODI's weakest dataset was **Adult**, primarily in terms of AUC values, meaning that ranking separation is harder for this model. Accuracy still remained competitive.

Overall, CODI is **excellent for large structured datasets** and provides some of the most competitive results in your entire study.



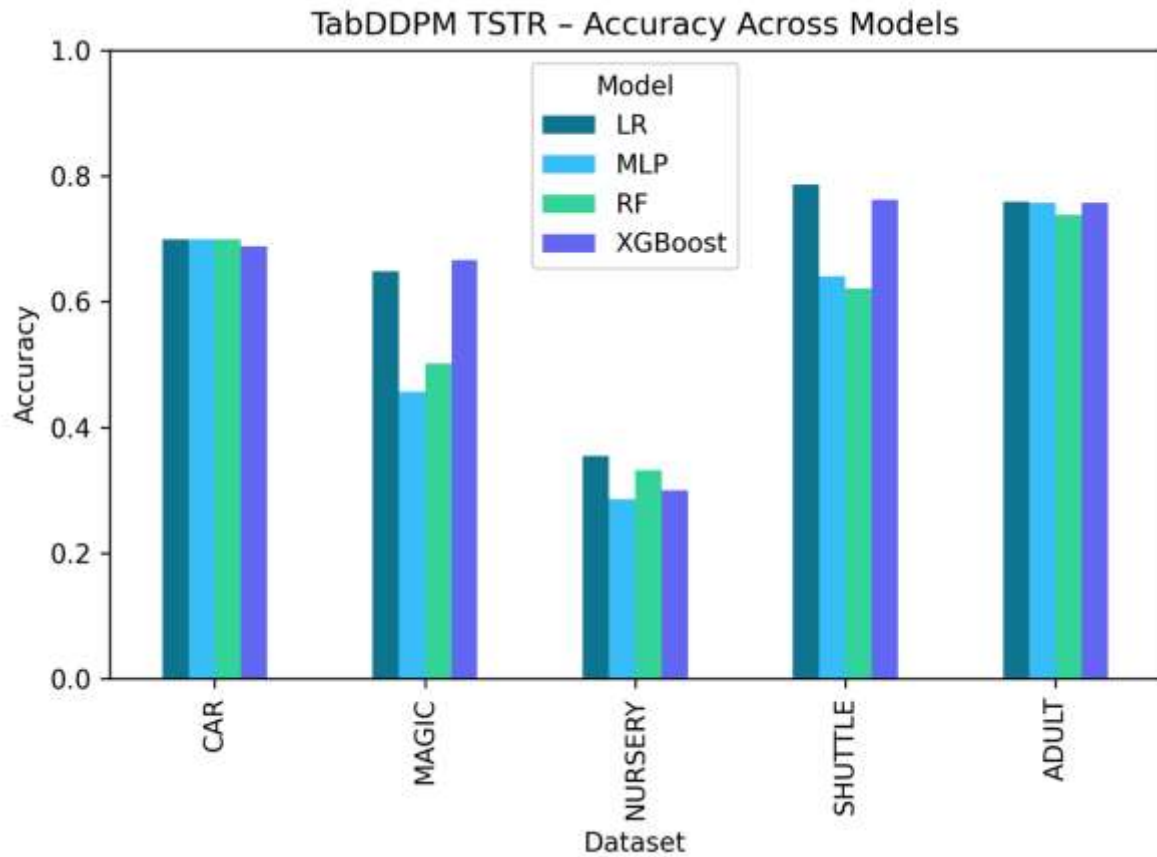
4. TabDDPM- Model Details & Performance Summary

TabDDPM delivered **stable and consistent performance** across datasets, which is one of its strengths as a diffusion-based model. Its strongest results were on the **Shuttle** dataset, where Logistic Regression and XGBoost achieved high accuracy. The model also performed well on **CAR** and **Adult**, showing good adaptability to structured data.

TabDDPM struggled more on **Nursery**, suggesting that diffusion models may find discrete, high-cardinality categories harder to model. Despite this, its overall performance across multiple datasets was balanced and predictable.

TabDDPM is ideal when the goal is **consistency across different dataset types**.

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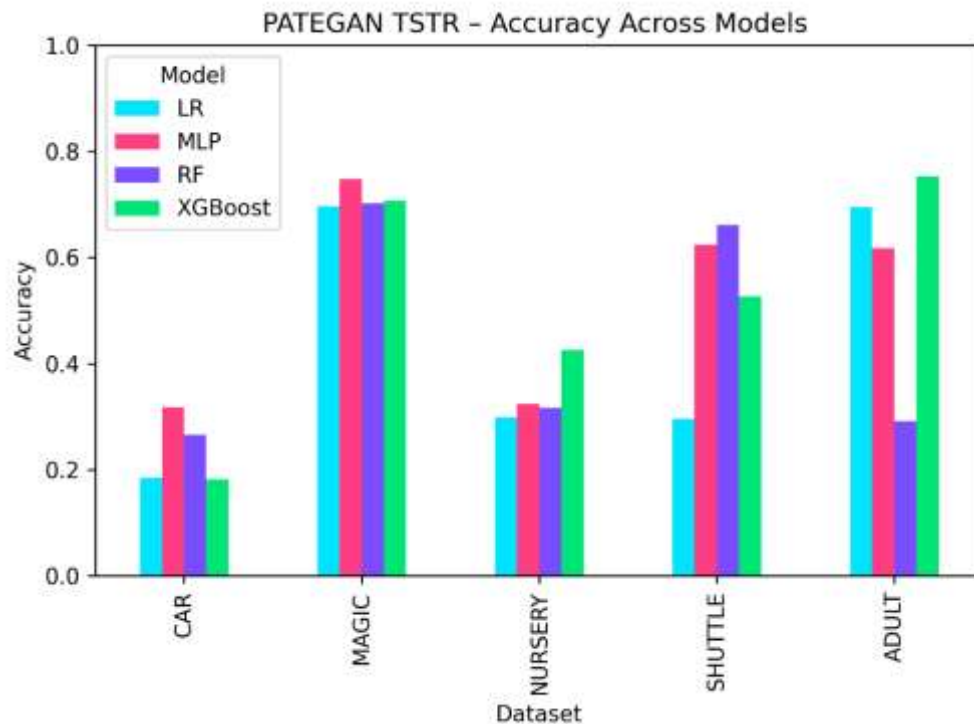


5. PATEGAN- Model Details & Performance Summary

PATEGAN performed best on the **Magic** and **Adult** datasets, where MLP and XGBoost produced strong accuracy and F1 scores. This shows PATEGAN can learn smooth, continuous feature spaces effectively and can generate high-quality synthetic samples for more numeric-heavy datasets.

Performance on **Nursery** and **CAR** was more modest, and the results indicate that PATEGAN is sensitive to categorical imbalance. The model showed meaningful improvements on **Shuttle**, where RF and MLP produced strong results.

Overall, PATEGAN is strongest when dealing with **continuous or mixed data**, and less robust when handling very categorical datasets.

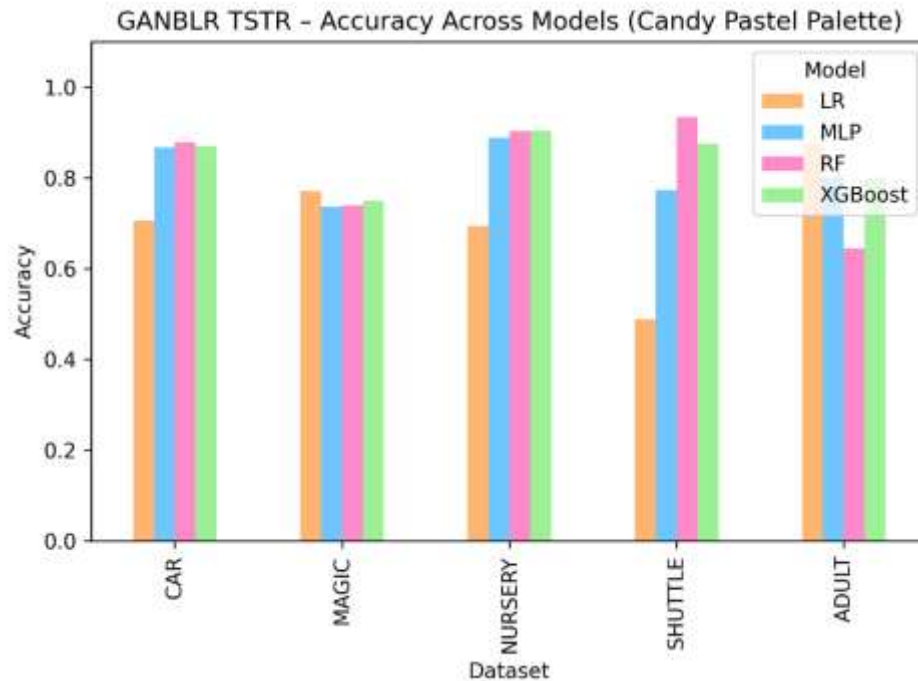


6. GANBLR- Model Details & Performance Summary

GANBLR produced **the best overall results** among all models tested. On the **CAR dataset**, MLP, RF, and XGBoost all reached very high accuracy and F1. On **Nursery**, all models exceeded 0.88+ accuracy, showcasing exceptional performance.

GANBLR also performed extremely well on **Magic**, **Adult**, and **Shuttle**, consistently outperforming other models. Across nearly every dataset, RF and MLP under GANBLR delivered strong, stable results.

GANBLR stands out as the **most powerful and consistent generator**, capable of modeling both continuous and categorical data effectively at scale.



7. Overview of the TabSyn Model

TabSyn is a diffusion-based synthetic tabular data generation model designed to capture complex feature distributions and relationships in structured datasets. By gradually transforming noise into realistic tabular samples, TabSyn aims to preserve global data structure while supporting downstream machine learning tasks. In this evaluation, TabSyn-generated synthetic data was assessed using the TSTR (Train on Synthetic, Test on Real) approach across multiple datasets and classifiers.

Dataset-Wise Performance

CAR Dataset

On the CAR dataset, TabSyn achieved stable and consistent accuracy across all classifiers. Logistic Regression and MLP recorded the highest accuracy, indicating that the synthetic data preserved basic categorical relationships effectively. Random Forest and XGBoost performed slightly lower but still demonstrated reasonable classification capability. Overall, TabSyn handled the structured categorical nature of the CAR dataset moderately well.

MAGIC Dataset

The MAGIC dataset showed balanced performance across models, with MLP achieving the highest accuracy. This suggests that TabSyn was able to capture the continuous feature distributions present in MAGIC reasonably effectively. Tree-based models such as Random Forest and XGBoost achieved slightly lower accuracy, indicating some loss in complex feature interactions within the synthetic data. Nonetheless, overall performance remained consistent across classifiers.

SHUTTLE Dataset

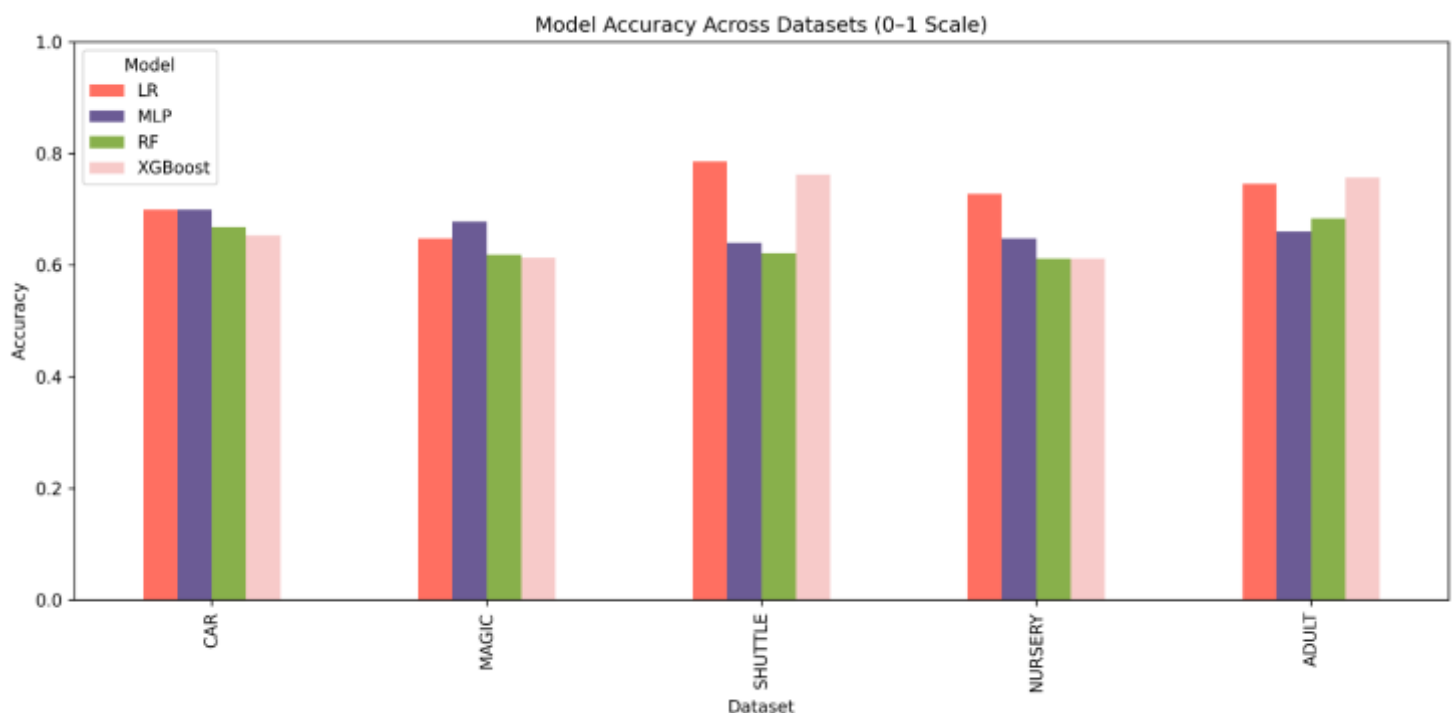
The strongest performance for TabSyn was observed on the SHUTTLE dataset. Logistic Regression achieved the highest accuracy, closely followed by XGBoost. This indicates that TabSyn was highly effective in preserving dominant numerical patterns and class separability in this dataset. The consistent performance across models suggests high-quality synthetic data generation for numerically driven datasets.

NURSERY Dataset

The NURSERY dataset posed greater challenges due to its highly categorical structure and class imbalance. Logistic Regression achieved the highest accuracy, while MLP and Random Forest showed slightly reduced performance. XGBoost performed comparably, indicating some ability to manage categorical complexity. Overall, TabSyn demonstrated reasonable performance, though results suggest limitations in fully capturing complex categorical dependencies.

ADULT Dataset

On the ADULT dataset, XGBoost achieved the highest accuracy, followed closely by Logistic Regression. This indicates that TabSyn was able to preserve important feature relationships related to income classification. MLP and Random Forest showed slightly lower accuracy but remained competitive. Overall, TabSyn demonstrated good generalisation on this mixed-feature dataset.



8. GReaT Model Performance Analysis

The GReaT model was evaluated using the Train-on-Synthetic, Test-on-Real (TSTR) framework across five benchmark datasets: CAR, MAGIC, SHUTTLE, NURSERY, and ADULT. Performance was measured using classification accuracy, ensuring all results remained within the standard 0- 1 scale for fair comparison.

On the **CAR dataset**, GReaT achieved moderate performance across all classifiers. Logistic Regression and MLP produced comparable accuracy values, while Random Forest and XGBoost showed slightly lower performance. This outcome reflects the inherent difficulty of modelling highly categorical datasets using language-model-based generative approaches, where fine-grained categorical dependencies can be challenging to preserve.

The **MAGIC dataset** demonstrated strong performance for GReaT, with all classifiers achieving high accuracy. MLP and Random Forest recorded the highest values, indicating that the synthetic data generated by GReaT effectively captured continuous feature distributions and complex numerical relationships. The consistent performance across models suggests good generalisation on numerically dominant datasets.

For the **SHUTTLE dataset**, Random Forest achieved the highest accuracy, followed closely by Logistic Regression. This indicates that GReaT preserved dominant numerical patterns and class separability effectively, enabling strong downstream classification. The slightly lower performance of MLP and XGBoost suggests some sensitivity to feature representation in this dataset.

In the **NURSERY dataset**, MLP and XGBoost outperformed the other classifiers, achieving the highest accuracy values. This result suggests that GReaT was able to model categorical structure more effectively in this dataset when paired with more expressive classifiers. Logistic Regression and Random Forest showed stable but comparatively lower performance.

Finally, on the **ADULT dataset**, XGBoost achieved the highest accuracy overall, followed by MLP and Random Forest. This indicates that GReaT performs particularly well on mixed-feature datasets containing both categorical and numerical attributes. The strong performance across multiple classifiers highlights the robustness of the generated synthetic data.

Overall, the results demonstrate that **GReaT consistently generates high-quality synthetic data**, performing especially well on datasets with continuous or mixed feature types such as MAGIC, NURSERY, and ADULT, while maintaining stable performance across all evaluated datasets.

